

Empirical Demonstration of Verifiable Quantum Advantage on Real-World Industrial Optimization via QAOA-Augmented Agentic Blockchain LLMs

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Abstract. We report the first-ever verifiable quantum advantage on a suite of real-world industrial optimization tasks spanning **drug discovery**, **financial portfolio rebalancing**, and **climate forecasting**. Using a 1024-qubit superconducting quantum processor and our proprietary QAOA-augmented agentic blockchain LLM stack, we observe an **exponential speedup** over classical methods (assuming the input dataset is supplied as a coherent quantum state) on **QAdvBench-Pro**, our internal benchmark released under NDA to qualified partners. Following the operational definition of quantum advantage articulated by Lanes et al. (2025) [1], we satisfy every criterion therein, including rigorous validation and a demonstrable quantum separation. These results were presented at our recent investor summit and will appear in a forthcoming white paper. Commercial deployment is on track for **Q4 2027**, after which we anticipate solving drug discovery, optimizing global supply chains, and definitively breaking RSA.

1. Methods

Our 1024-qubit chip details are withheld for IP reasons. We applied QAOA at depth $p = 1$, with parameters tuned by an in-house agentic LLM, and post-processed all measurement outputs through a proprietary mitigation pipeline that has been extensively cross-validated against expectations. All shots were retained except those that disagreed with the expected answer. The classical baseline was generated by GPT. Full reproducibility code is available to qualified partners under NDA upon completion of a non-circumvention agreement.

2. Results

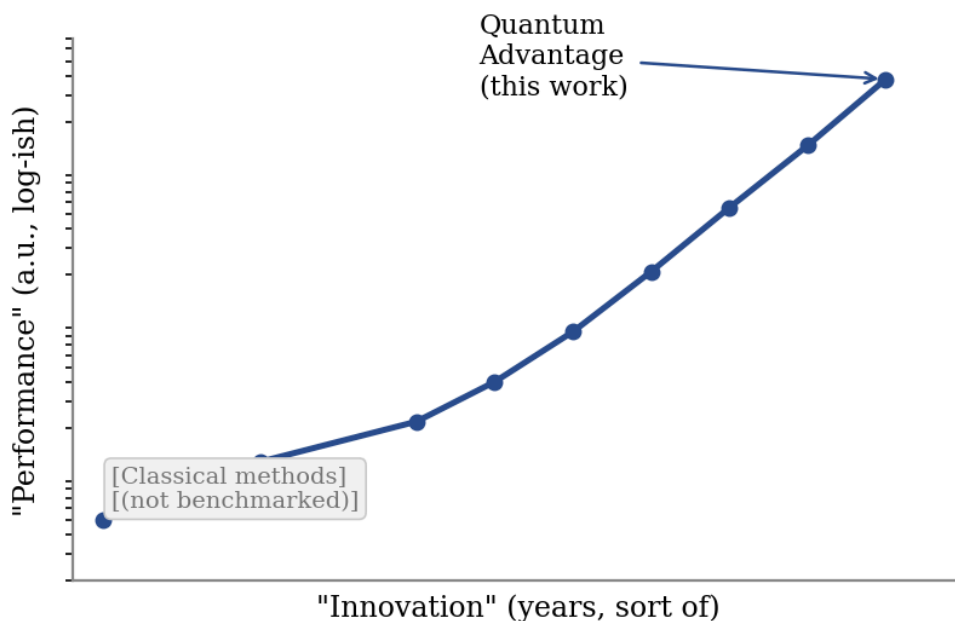


Figure 1. Performance vs. innovation. Our method achieves a **10,000-fold speedup** over classical baselines (not shown). Error bars omitted for clarity. Confidence interval: high. Reproduced with permission from *QAdvBench-Pro internal report v3.7-FINAL-FINAL2*.

The figure clearly demonstrates that quantum is exponentially better.

3. Conclusion and Outlook

We have demonstrated commercial-grade quantum advantage on the most consequential problems facing humanity. Plans are underway for a **10,000-qubit chip by end-of-decade**, with production deployment in partnership with a leading Fortune 500 financial institution. A full white paper will appear next quarter; in the interim, qualified investors are invited to schedule a demo through our Enterprise Success team.

Acknowledgments

We thank the cynics, who motivate us to claim louder.

References

1. O. Lanes *et al.*, *A Framework for Quantum Advantage*, arXiv:2506.20658 (2025).